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SPID rotors, like the RAS and BIG RAS, have a large nut on the end of the worm gear and a notice "Do Not Regulate". You probably want to know what the purpose of this nut and adjustment is ? Looks like it must be holding one end of the worm gear, but what is it adjusting? How hard the worm gear is gripped ?



Answering to this question here.....

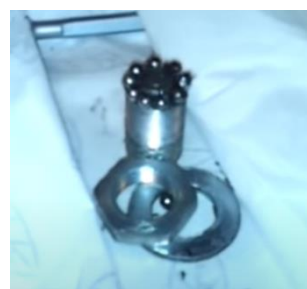
Once you loosen the large nut, the screw with the large slot adjusts the preload on an open race ball bearing which holds one end of the worm gear. As with all open race ball bearing adjustment, it should be snug, but not tight. If you crank it down tight, the balls will not roll freely. Best way to set it is to remove the motor and rotate the worm gear by hand (long nose pliers or make a special tool), then see how it feels as you rotate it. If it feels "gritty", you have it too tight. Just before it gets "gritty" is the right spot.

The hole through which this large screw passes is elongated. When the screw is at the top of the slot, the clearance between the worm and the ring gear is at its minimum, i.e. in close permanent contact. This is not desirable and will lead to wear, loading on the motor and possibly "grinding noises". When the screw is at the bottom of the slot, there will be noticeable play (backlash) in the main gear (and output shaft). Again, not desirable. The ideal position would be just barely detectable play (backlash). This means the gears are not in constant close contact. Operation will be smoother and wear will be lower. The purpose of the large external nut is to lock everything in place once the preload and backlash clearance are set.

If you back the screw all the way out the ball bearings will fall out. If you don't want to replace these ball bearings don't back out the screw too far.

Why should you not regulate ?

SPID Rotators do not use ball bearings in case of the worm gear - there are only balls (each side). The two large nuts (1st in and 2nd out) hold the screw which clamps the worm. As particular rotator's parts may slightly differ (due to many reasons) SPID have to sort them to have optimal match of the worm wheel with the worm (it refers to each our device). The warning sticker is to avoid adjustment by an unaware user (as this may result in inappropriate rotator's work).



Note that this screw is marked "DO NOT REGULATE", so if you mess with it, do so at your own risk. It's probably best to leave it alone.

In General:

If you do have problems with this gear, contact us and we can offer you a proposal for repair.